Inverse and Inverse-Trig Derivatives
Answer Key
AP Calculus AB/BC

Quick Answers

1. $\frac{3}{\sqrt{1-9x^2}}$
2. $\frac{2x}{1+x^4}$
3. (a) the derivative of f = $3x^2+2$, (b) the derivative of the inverse at 4 = $\frac{1}{5}$
4. $\operatorname{atan}(x) + \frac{x}{1+x^2}$
5. $-\frac{2}{\sqrt{1-4x^2}}$
6. (a) the derivative of g = $5x^4+3$, (b) the derivative of the inverse at 2 = $\frac{1}{8}$
7. $\frac{2\operatorname{atan}(x)}{1+x^2}$
8. $\frac{1}{2\sqrt{x}(1+x)}$
9. $2x\operatorname{acos}(x) - \frac{x^2}{\sqrt{1-x^2}}$
10. (a) the derivative of f = $2+\cos(x)$, (b) the derivative of the inverse at 0 = $\frac{1}{3}$, —

What is verified

13 of 14 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC
FUN-3.E – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

3. *If they got 50: evaluated the derivative of f at the output 4 instead of taking the reciprocal at the input 1*
6. *If they got 0.5: took the reciprocal of the input value 2 instead of the reciprocal of the derivative*

1. Differentiate $y = \arcsin(3x)$.
Solution: With $u = 3x$ we have $u' = 3$, and $\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$:

$$y' = \frac{3}{\sqrt{1-(3x)^2}} = \frac{3}{\sqrt{1-9x^2}}. \qquad \boxed{\frac{3}{\sqrt{1-9x^2}}}$$

The 3 upstairs is the chain-rule factor; squaring $3x$ puts a 9 downstairs.

2. Differentiate $y = \arctan(x^2)$.
Solution: With $u = x^2$, $u' = 2x$, and $\frac{d}{dx}\arctan u = \frac{u'}{1+u^2}$:

$$y' = \frac{2x}{1+(x^2)^2} = \frac{2x}{1+x^4}. \qquad \boxed{\frac{2x}{1+x^4}}$$

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3. $f(x) = x^3 + 2x + 1$ with $f(1) = 4$. (a) $f'(x)$. (b) $\left(f^{-1}\right)'(4)$.
Solution (a): Power rule term by term: $\boxed{3x^2 + 2}$
Solution (b): The rule is $\left(f^{-1}\right)'(b) = \dfrac{1}{f'(a)}$ where $f(a) = b$. Here $b = 4$ and $a = 1$, so evaluate f' at the *input* 1, not at 4:

$$f'(1) = 3(1)^2 + 2 = 5 \qquad \Longrightarrow \qquad \left(f^{-1}\right)'(4) = \frac{1}{5}.$$

So the answer is $\boxed{\frac{1}{5}}$

Using $f'(4) = 50$ instead is the standard slip; 50 is the number to watch for on a wrong page.

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4. Differentiate $y = x \arctan x$.
Solution: Product rule with $u = x$ and $v = \arctan x$:

$$y' = (1)\arctan x + x \cdot \frac{1}{1+x^2} = \arctan x + \frac{x}{1+x^2}.$$

So the derivative is $\boxed{\arctan x + \dfrac{x}{1+x^2}}$

The answer keeps an arctan in it; that is normal and not a sign of an error.

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5. Differentiate $y = \arccos(2x)$.
Solution: With $u = 2x$, $u' = 2$, and the arccosine rule carrying the minus sign:

$$y' = \frac{-2}{\sqrt{1-(2x)^2}} = \frac{-2}{\sqrt{1-4x^2}}. \qquad \boxed{\frac{-2}{\sqrt{1-4x^2}}}$$

Arcsine and arccosine have the same derivative up to sign, because their sum is the constant $\frac{\pi}{2}$.

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6. $g(x) = x^5 + 3x - 2$ with $g(1) = 2$. (a) $g'(x)$. (b) $\left(g^{-1}\right)'(2)$.
Solution (a): $\boxed{5x^4 + 3}$
Solution (b): The input that produces the output 2 is $a = 1$, so

$$g'(1) = 5 + 3 = 8 \qquad \Longrightarrow \qquad \left(g^{-1}\right)'(2) = \frac{1}{8}.$$

So the answer is $\boxed{\frac{1}{8}}$

Note it is the *derivative* that gets reciprocated, not the input: $\frac{1}{2}$ is the wrong answer this problem is built to catch.

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7. Differentiate $y = (\arctan x)^2$.

Solution: Chain rule with the outside function $(\ )^2$ and the inside function $\arctan x$:

$$y' = 2\arctan x \cdot \frac{1}{1+x^2} = \frac{2\arctan x}{1+x^2}. \qquad \boxed{\frac{2\arctan x}{1+x^2}}$$

Contrast with problem 2, where arctan was the *outside* function — same two ingredients, opposite order, different answer.

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8. Differentiate $y = \arctan(\sqrt{x})$.

Solution: With $u = \sqrt{x}$ we have $u' = \frac{1}{2\sqrt{x}}$ and $u^2 = x$, so

$$y' = \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}\,(1+x)}. \qquad \boxed{\frac{1}{2\sqrt{x}\,(1+x)}}$$

Squaring $\sqrt{x}$ back to x is what makes this one simplify so cleanly.

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9. Differentiate $y = x^2 \arccos x$.

Solution: Product rule with $u = x^2$ and $v = \arccos x$:

$$y' = 2x\arccos x + x^2\left(\frac{-1}{\sqrt{1-x^2}}\right) = 2x\arccos x - \frac{x^2}{\sqrt{1-x^2}}.$$

So the derivative is $\boxed{2x\arccos x - \frac{x^2}{\sqrt{1-x^2}}}$

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10. Challenge. $f(x) = 2x + \sin x$, $f(0) = 0$. (a) $f'(x)$. (b) $(f^{-1})'(0)$. (c) Why does the inverse exist?

Solution (a): $\boxed{2 + \cos x}$

Solution (b): The input producing the output 0 is $a = 0$, and $f'(0) = 2 + \cos 0 = 2 + 1 = 3$, so $(f^{-1})'(0) = \frac{1}{3}$. $\boxed{\frac{1}{3}}$

Notice that f^{-1} has no formula here — there is no way to solve $y = 2x + \sin x$ for x — and the rule still delivers the slope. That is the point of the inverse-function rule.

Solution (c) — instructor-judged, no machine check. The argument must use part (a). Cosine never falls below -1, so $f'(x) = 2 + \cos x \geq 1 > 0$ for every x. A function whose derivative is positive everywhere is strictly increasing, and a strictly increasing function takes each value at most once, so it is one-to-one and has an inverse on all of $\mathbb{R}$. Full credit needs the positivity of the derivative *and* the step from strictly increasing to one-to-one. Do not accept "it passes the horizontal line test" with no reason given for why it does.